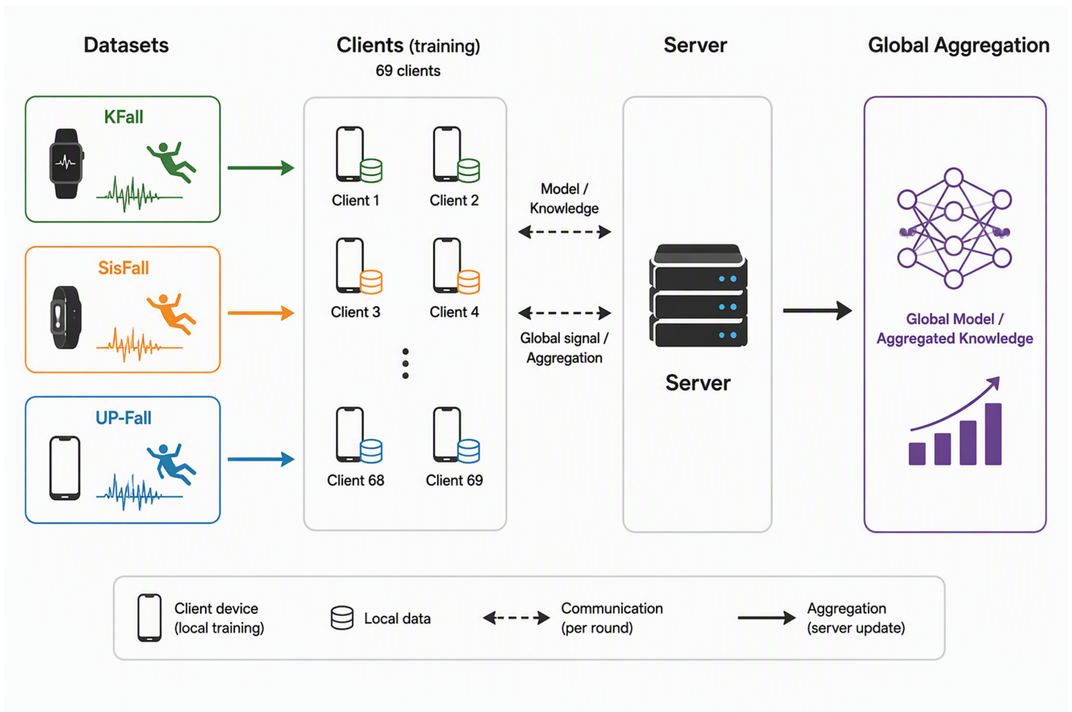
Revisiting Communication in Federated Learning: Parameter vs. Knowledge-Based Strategies

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Motivation & Research Goals

Federated Learning (FL) enables collaborative model training without sharing raw data, making it particularly suitable for privacy-sensitive healthcare applications. However, performance strongly depends on the communication mechanism used between clients and the server, especially under heterogeneous and non-IID data distributions. This work compares parameter-sharing and knowledge-based communication strategies for wearable fall detection, investigating their impact on predictive performance and communication efficiency.

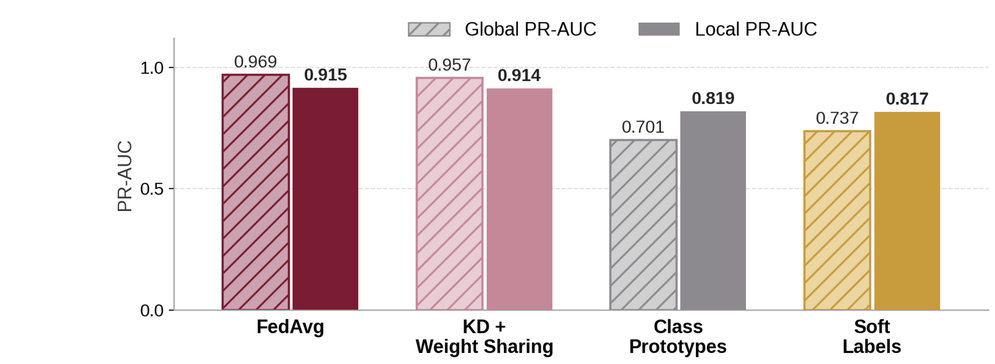
Methods



Datasets: KFall, SisFall and UP-Fall
Experimental Setup: 69 Training Clients, 9 Development Clients and 9 Test Clients
Model : $32 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 1$

Strategy	Transferred Information
FedAvg	Model weights
KD + Weight Sharing	Model weights + distillation
Soft Labels	Prediction statistics
Class Prototypes	Class-level representations

Results



	FedAvg	KD + Weight Sharing	Class Prototypes	Soft Labels
Values exchanged per round	49,665	49,665	131	2
Reduction vs FedAvg	–	0 %	99.74 %	99.996 %

Main Findings

- **99.996 %** — Minor communication with soft labels, Local PR-AUC decreases by only ≈0.1
- **≈ 65×** — More data with prototypes for only +0.002 PR-AUC
- **Δ −0.001** — Knowledge Distillation: limited gain over FedAvg
- **Local > Global** — Knowledge-sharing scores higher locally than globally

Conclusion

FedAvg achieved the best global performance under heterogeneous non-IID data. Knowledge-based communication strategies achieved lower global performance but maintained competitive local performance while dramatically reducing communication costs. These results suggest that lightweight knowledge-sharing mechanisms constitute practical alternatives for communication-constrained federated healthcare applications.

Key References

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For the complete paper, supplementary material, and full bibliography, please scan the QR code.

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